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Next item →

1. Let $\mathcal{R}$ be a binary relation on a set A , Which one of the following is true about the relation $\mathcal{R}$? 1 / 1 point
- ☐ $\mathcal{R}$ is transitive if and only if $\forall x, y \in A$, if $x\mathcal{R}y$ and $y\mathcal{R}x$ then $x = y$

☐ $\mathcal{R}$ is transitive if and only if $\forall x, y \in A$, if $x\mathcal{R}y$ then $y\mathcal{R}x$

☐ $\mathcal{R}$ is transitive if and only if $\forall x \in A, x\mathcal{R}x$

☒ $\mathcal{R}$ is transitive if and only if $\forall x, y, z \in A$, if $x\mathcal{R}y$ and $y\mathcal{R}z$ then $x\mathcal{R}z$

✔ Nice work

Correct. This is the definition of a transitive relation.

2. Let $\mathcal{R}$, a binary relation on $\mathbb{R}$ be defined as $a\mathcal{R}b$ if and only if $a < b, \forall a, b \in \mathbb{R}$. Which one of the following statements is true about the relation $\mathcal{R}$? 1 / 1 point
- ☒ $\mathcal{R}$ is transitive and anti-symmetric

☐ $\mathcal{R}$ is anti-symmetric but not transitive

☐ $\mathcal{R}$ is transitive and reflexive

☐ $\mathcal{R}$ is transitive and symmetric

✔ Nice work

Correct. It can be shown that $\mathcal{R}$ is transitive and anti-symmetric. However, $\mathcal{R}$ is not reflexive as $x \not\mathcal{R} x$ and not symmetric as $2 < 3$ but $3 \not< 2$.

3. The binary relation $R = \{(0, 0), (1, 1)\}$ on elements of a set $A = \{0, 1, 2, 3\}$ is: 1 / 1 point
- ☐ symmetric, reflexive and transitive

☒ symmetric and transitive but not reflexive

☐ reflexive, anti-symmetric and transitive

☐ reflexive but neither symmetric nor transitive

✔ Nice work

Correct. The relation is not reflexive as $2 \not\mathcal{R} 2$ or $3 \not\mathcal{R} 3$.

4. Let $A = \{a, b, c, d\}$ and $\mathcal{R}$ be a binary relation on A defined by $\mathcal{R} = \{(a, b), (b, c), (c, d)\}$. What is the transitive closure of the relation $\mathcal{R}$? 1 / 1 point
- ☐ $\{(a, b), (b, c), (a, c), (b, d), (a, d)\}$

☐ $\{(a, b), (b, c), (a, c), (c, d), (a, d)\}$

☒ $\{(a, b), (b, c), (a, c), (c, d), (b, d), (a, d)\}$

☐ $\{(a, b), (b, c), (a, c), (c, d), (b, d)\}$

✔ Nice work

Correct. $\mathcal{R}$ is not transitive; the transitive closure of the relation $\mathcal{R}$ is obtained by adding the minimum number of connections to make the relation transitive.